

20 June 2026

Theoretical Developments in Gravitational Wave Physics

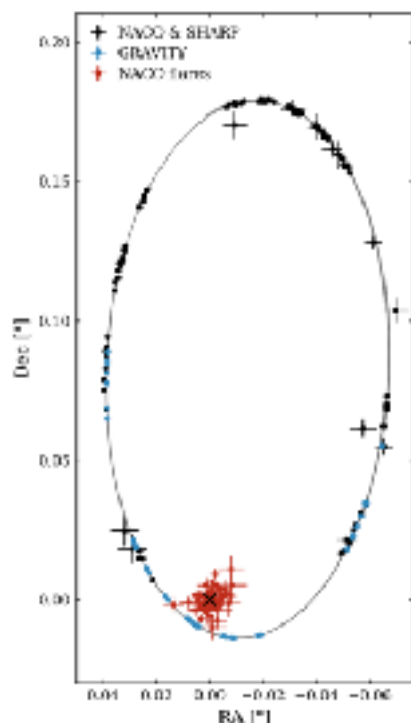
Ringdown tests of the black-hole paradigm

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INFN Rome (Italy)

Tests of general relativity

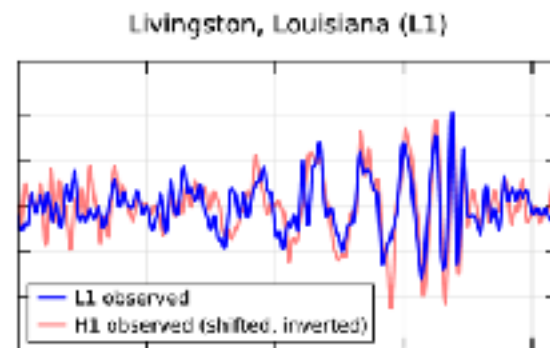
Current electromagnetic and gravitational observations have probed general relativity and the spacetime of black holes.



GRAVITY, A&A **636**, L5 (2020)



EHT, ApJL **910**, L12 (2021)



Abbott+, PRL **116**, 061102 (2016)

Black holes in general relativity

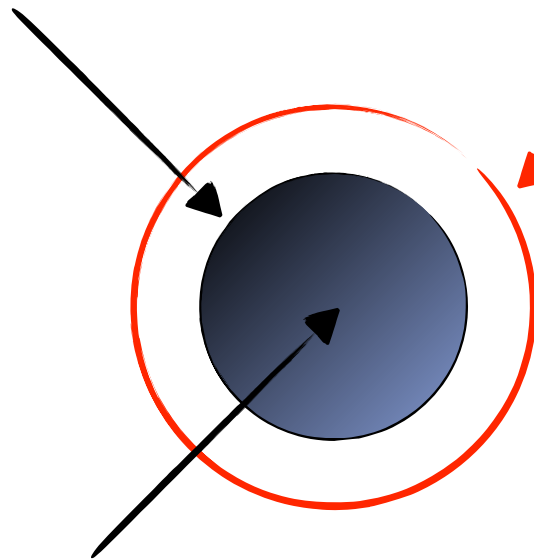
The black-hole spacetime has been probed approximately until the location of the light ring.

Horizon $r_H = 2\frac{GM}{c^2}$

The curvature changes
the causal structure

Light ring $r = 3\frac{GM}{c^2}$

Unstable circular
orbit of photons



Curvature singularity $r = 0$

Breakdown of Einstein equations

Black hole paradoxes

- Curvature singularity at $r = 0$
- Closed timelike curves violate causality inside the horizon

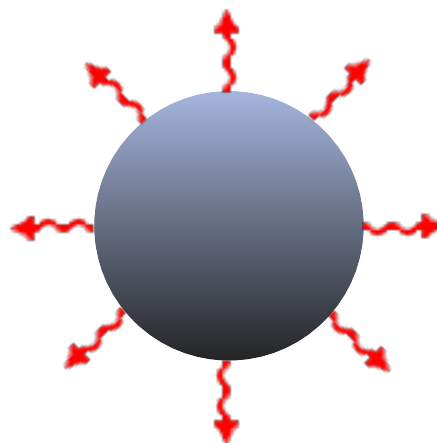
- Thermodynamical instability due to Hawking radiation at $T_H = \frac{\hbar c^3}{8\pi k_B G M}$

Hawking, PRD **13**, 191 (1976)

- Huge entropy $S_{\text{BH}} = \frac{4\pi k_B G M^2}{\hbar c}$

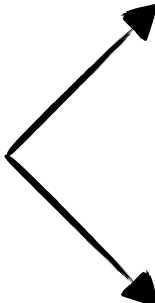
Bekenstein, PRD **7**, 2333 (1973)

- Information-loss paradox



Horizonless compact objects

New physics can prevent the formation of the horizon and the singularity:



In **extensions of general relativity**

(e.g. fuzzballs Mathur, Fortsch. Phys. **53**, 793-827 (2005))

In general relativity with **dark matter or exotic fields**

(e.g. boson stars Liebling+, LRR **20**, 5 (2017))

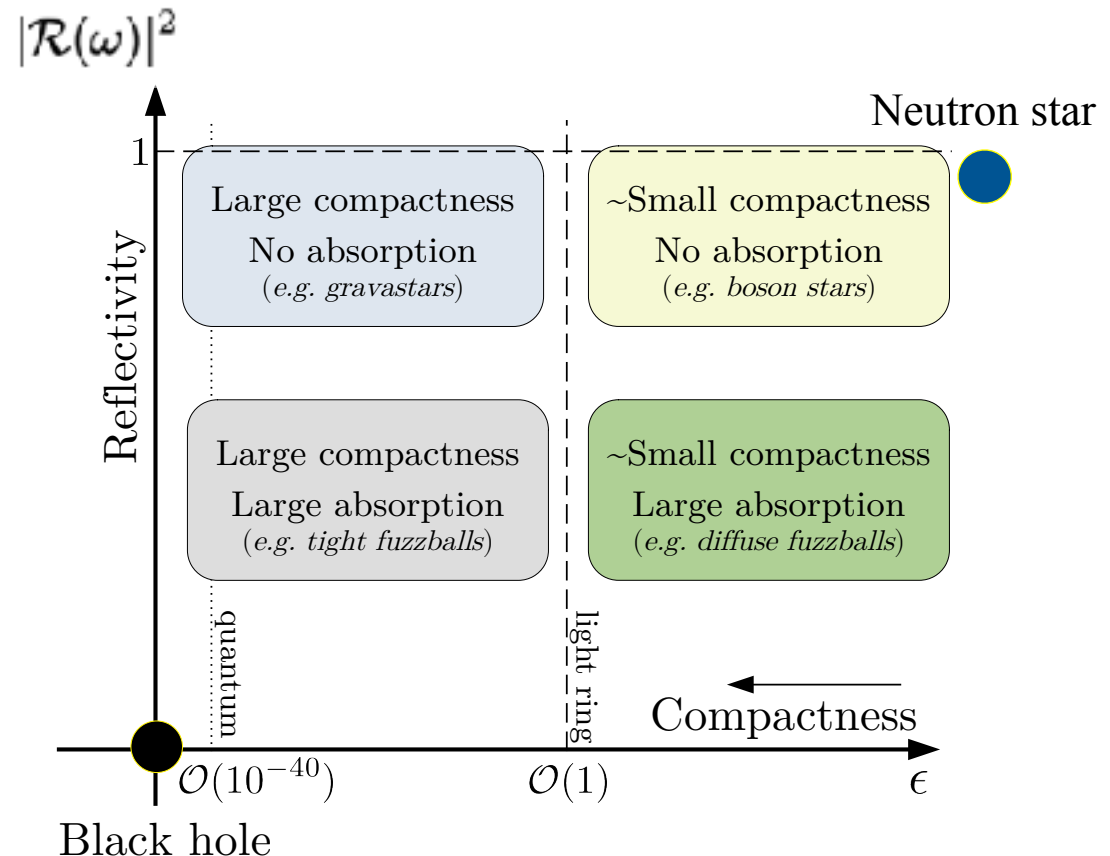
Horizonless compact objects can mimic black holes and quantify the existence of horizons.

Giudice+, JCAP **10** (2010) 001; Cardoso+, LRR **22**:4 (2019); EM+, Handbook for GW Astronomy, Springer (2021)

A parametrised classification

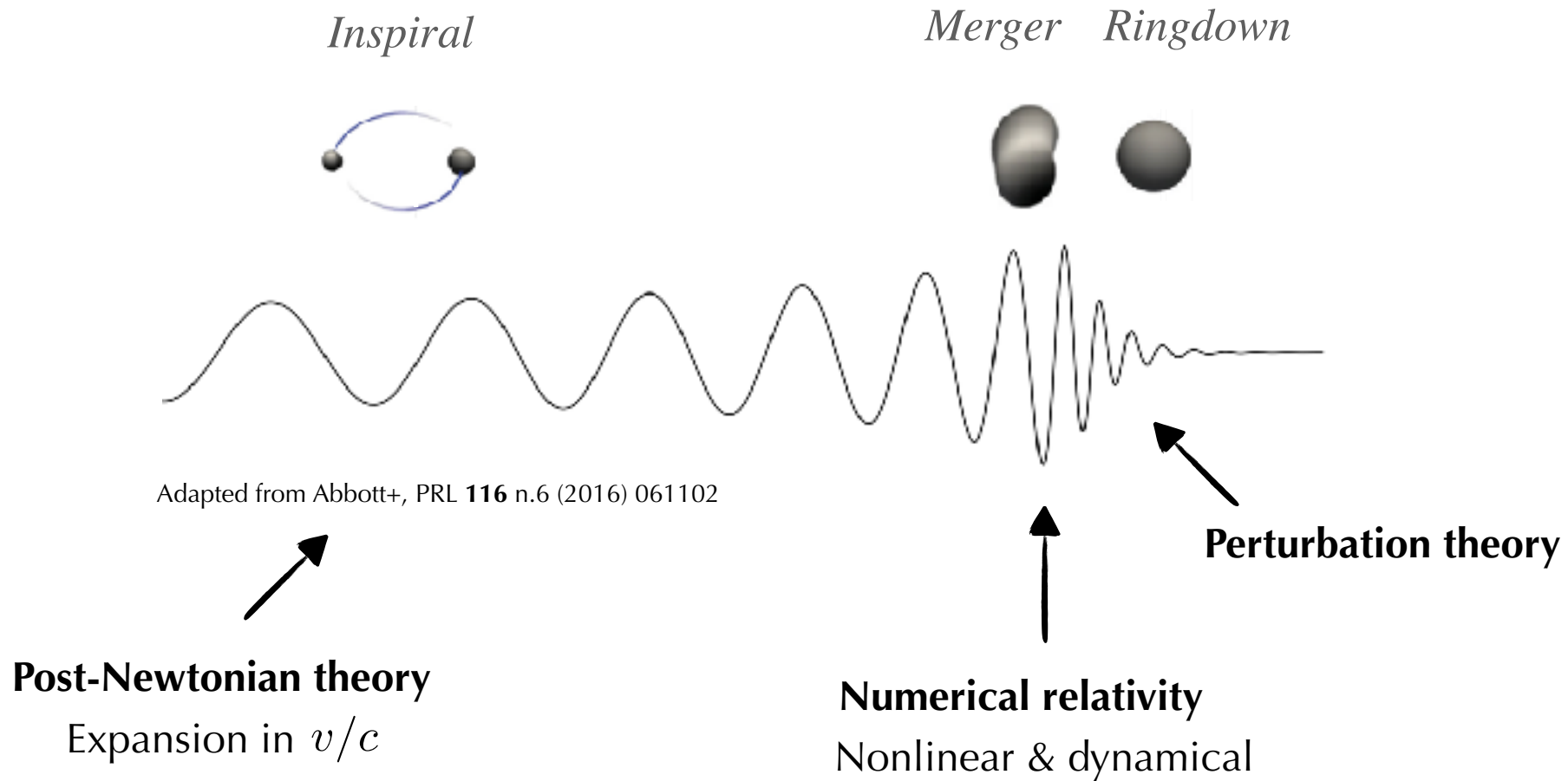
A horizonless compact object deviates from a black hole for its:

- **Compactness**
since the object's radius is at
 $r_0 = r_H(1 + \epsilon)$
- **Reflectivity**
Unlike a totally absorbing black hole



EM, Pani, Raposo, Handbook for GW Astronomy, Springer (2021)

Gravitational waves from binary mergers



Modeling gravitational waves

For horizonless compact objects:

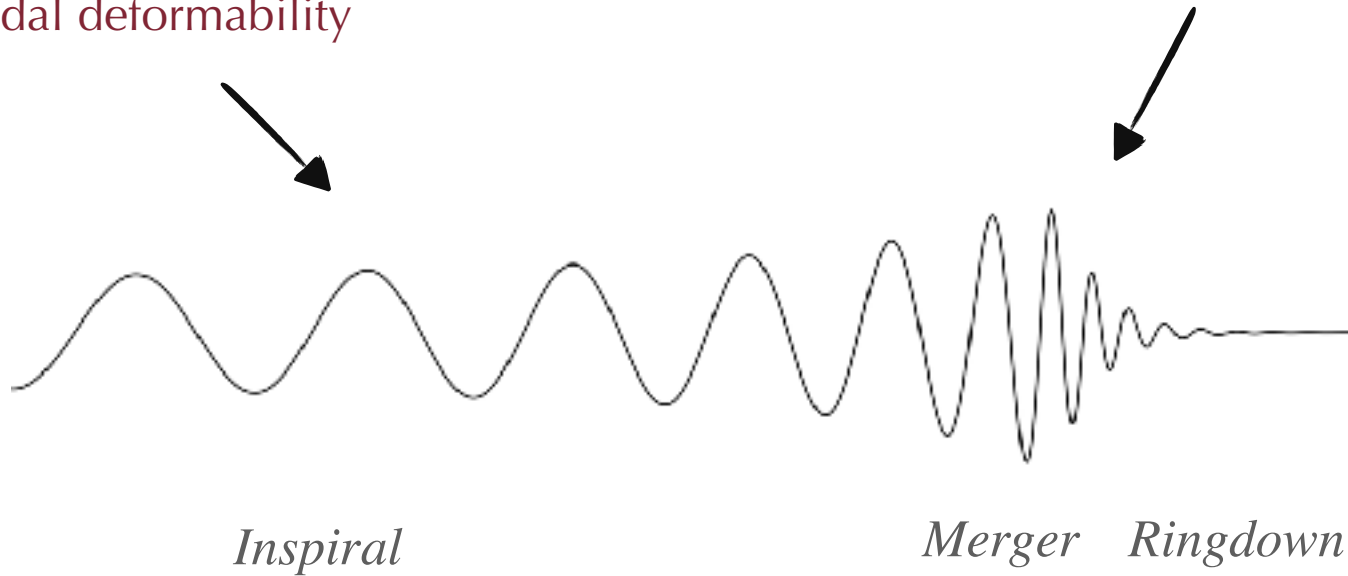
Spin-induced multipoles

Tidal heating

Tidal deformability

Modified spectrum

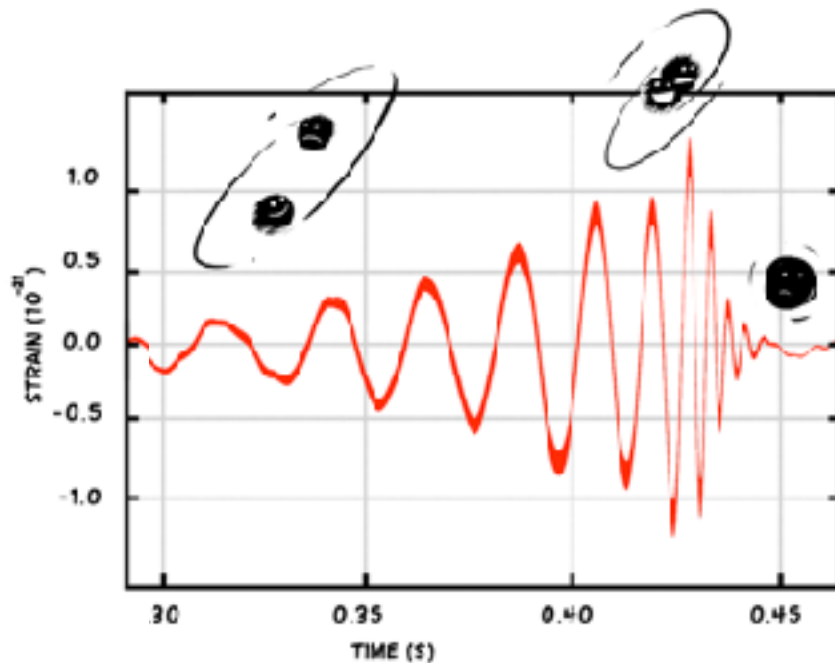
Echoes



The ringdown

The ringdown is dominated by the characteristic oscillation frequencies of the remnant, known as **quasi-normal modes**:

$$\omega_{lmn} = \omega_{R,lmn} + i\omega_{I,lmn}$$



It is modeled as a sum of exponentially damped sinusoids:

$$f_{lmn} = \frac{\omega_{R,lmn}}{2\pi}$$

$$\tau_{lmn} = -\frac{1}{\omega_{I,lmn}}$$

<https://www.ligo.org/magazine/LIGO-magazine-issue-8.pdf>

The quasinormal modes

We analyze a horizonless compact object described by the Schwarzschild metric and perturbed by a gravitational perturbation:

$$\frac{d^2\psi(r)}{dr_*^2} + [\omega^2 - V(r)] \psi(r) = 0$$

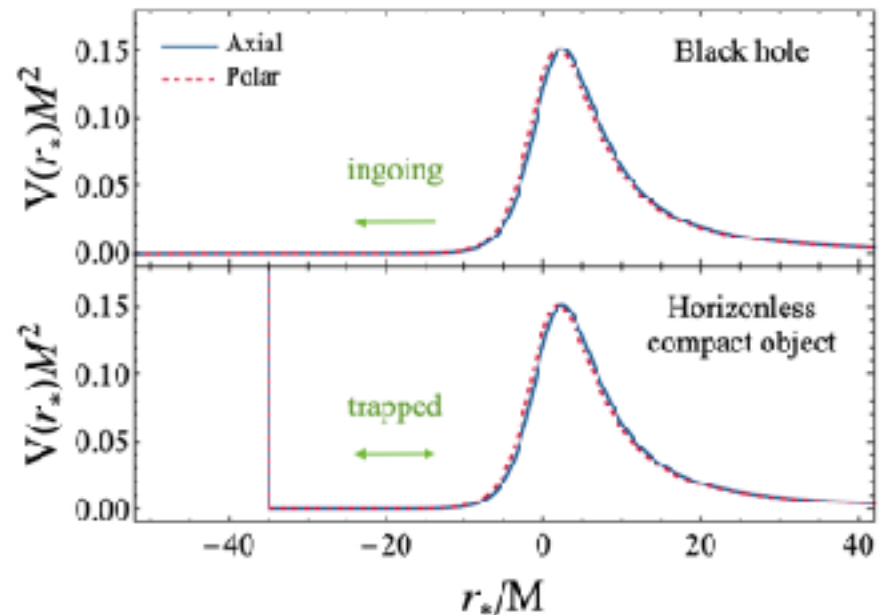
Regge, Wheeler, Phys.Rev. **108** (1957) 1063-1069

Zerilli, PRL **24** (1970) 737-738

+ 2 boundary conditions:

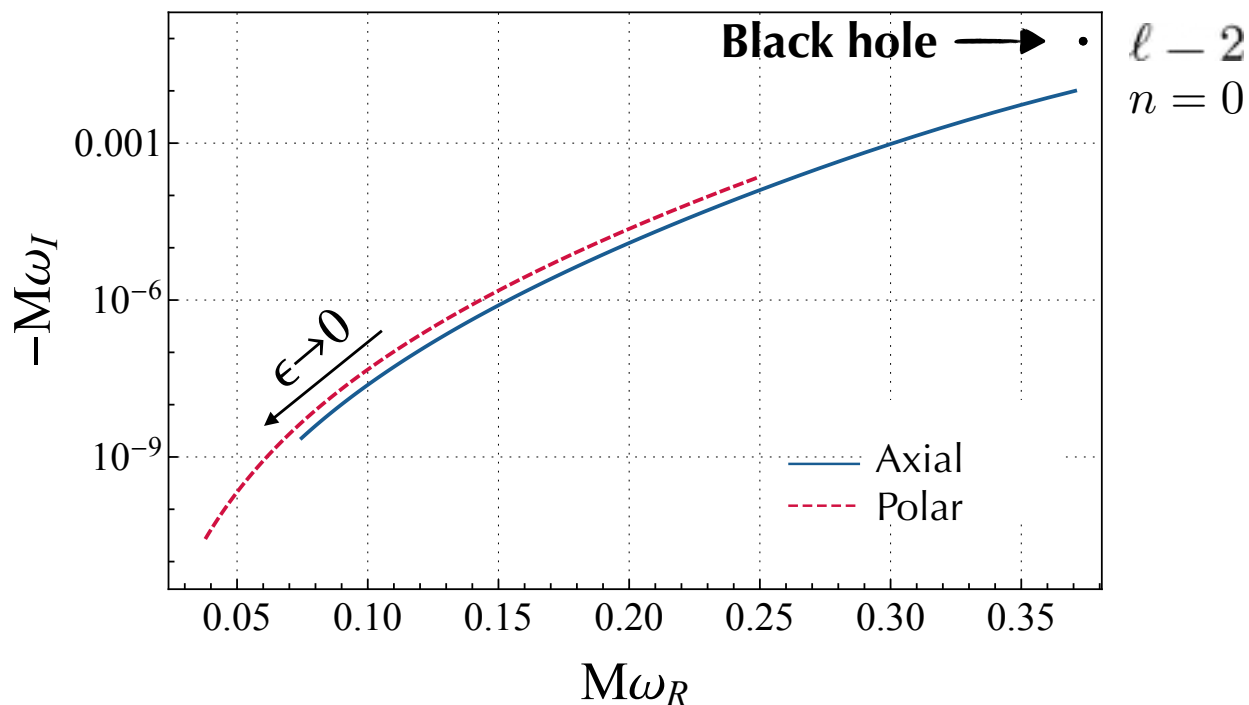
- At infinity: outgoing waves
- At the object's radius: $\psi(r_0) \sim e^{-i\omega r_*} + \mathcal{R}(\omega)e^{i\omega r_*}$
for ultracompact objects ($\epsilon \ll 1$)

EM, Pani, Raposo, Handbook for GW Astronomy, Springer (2021)



Quasi-normal mode spectrum

Ultracompact and reflective object ($\epsilon \ll 1$):



- Axial and polar modes are not isospectral.
- For $\epsilon \rightarrow 0$, the quasi-normal modes are low-frequencies and long-lived.

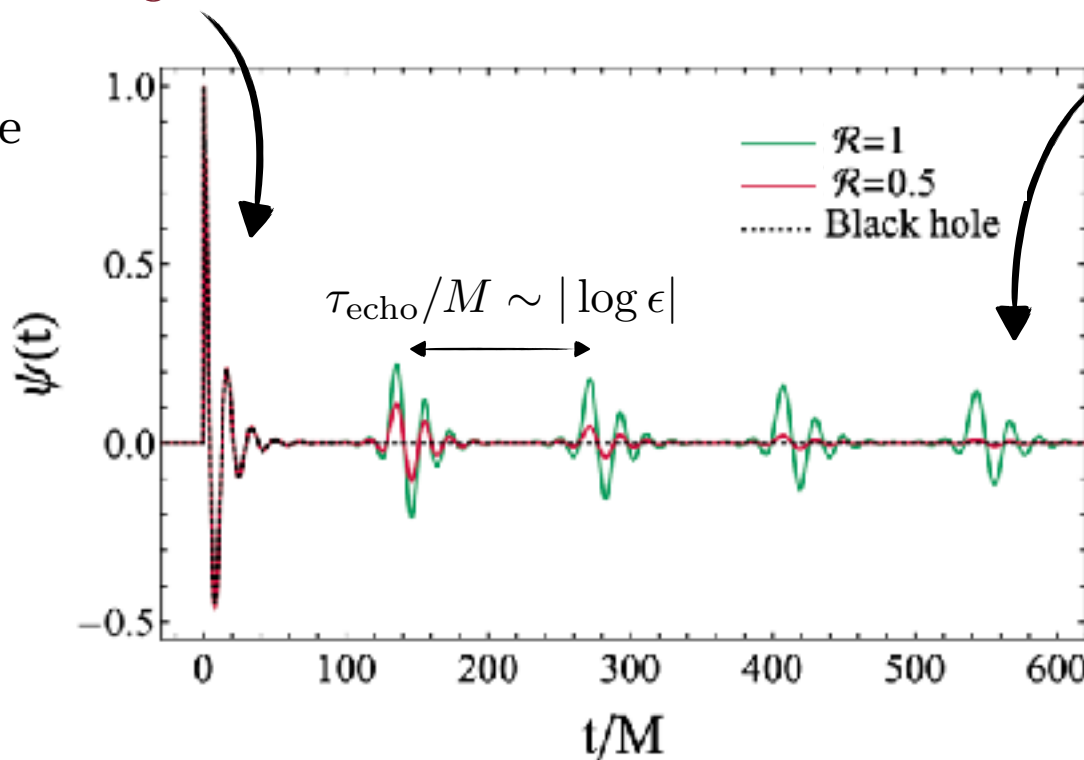
Cardoso+, PRL **116**, 171101 (2016); EM, Pani, Raposo, Handbook for GW Astronomy, Springer (2021)

Ringdown of horizonless objects

Ultracompact and reflective object ($\epsilon \ll 1$):

Same black hole ringdown

due to the
excitation of the
light ring



Gravitational-wave
echoes

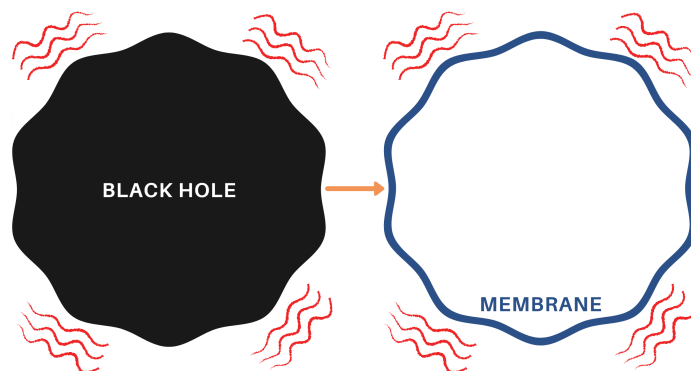
due to trapped
modes

Cardoso+, PRL **116**, 171101 (2016); EM, Testa, Bhagwat, Pani, PRD **100**, 064056 (2019);
EM, Pani, Raposo, Handbook for GW Astronomy, Springer (2021)

Membrane paradigm

We derive the boundary condition that describes **horizonless compact objects with any compactness** and a Schwarzschild exterior.

EM, Buoninfante, Mazumdar, Pani, PRD **102**, 064053 (2020)



Damour, PRD **18**, 10 (1978); Price, Thorne, PRD **33**, 4 (1986)

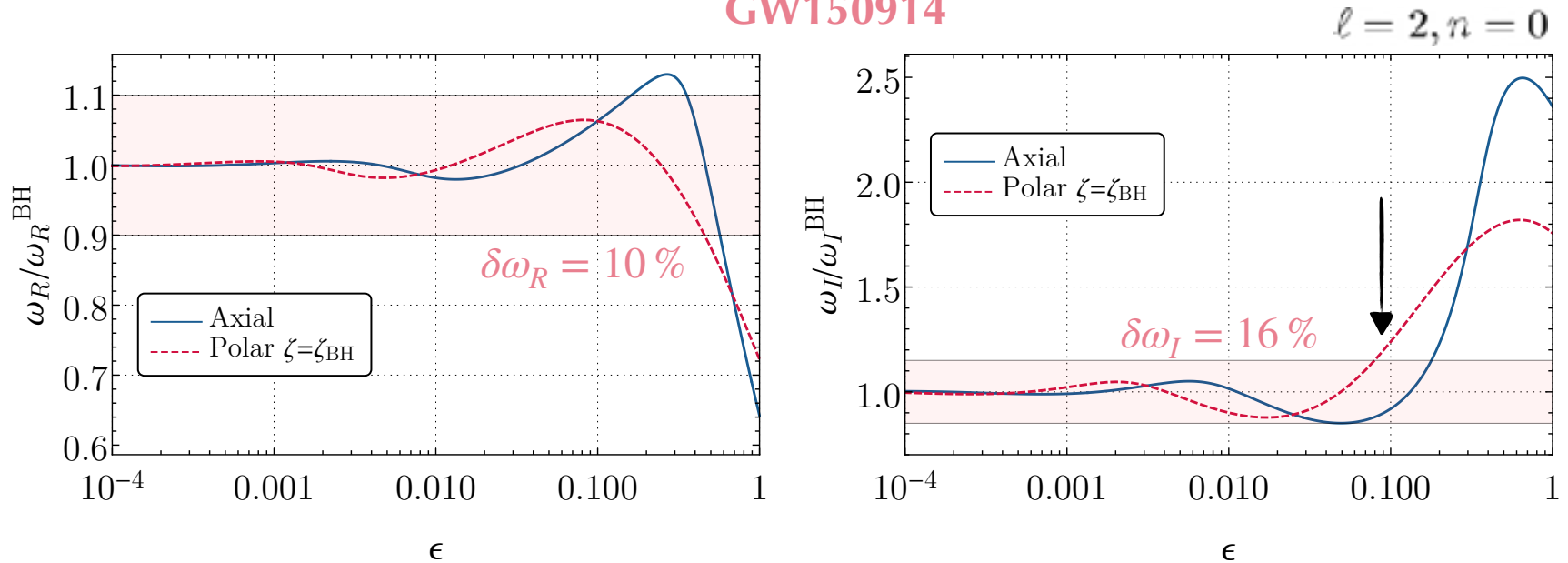
A static observer can replace the interior of a black hole by a **fictitious** membrane located at the horizon, which is a **viscous fluid** with shear viscosity η and bulk viscosity ζ .

Spectrum of compact objects

Totally absorbing compact object with $\epsilon \gtrsim 0.01$:

$$r_0 = r_H(1 + \epsilon)$$

GW150914



EM, Buoninfante, Mazumdar, Pani, PRD **102**, 064053 (2020)

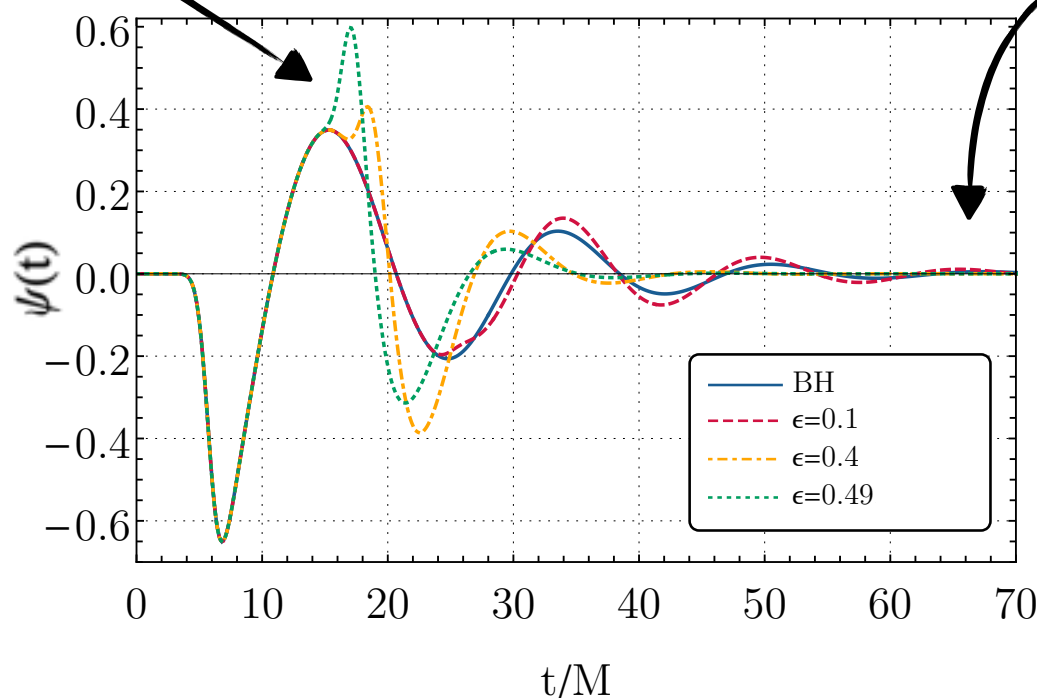
The measurement precision of the fundamental quasinormal mode of GW150914 excludes horizonless compact objects with $\epsilon \gtrsim 0.1$.

Ringdown of horizonless objects

Totally absorbing compact object with $\epsilon \gtrsim 0.01$:

Modified ringdown

due to the
interference with
the first echo



No echoes

the cavity does
not trap efficiently
the modes

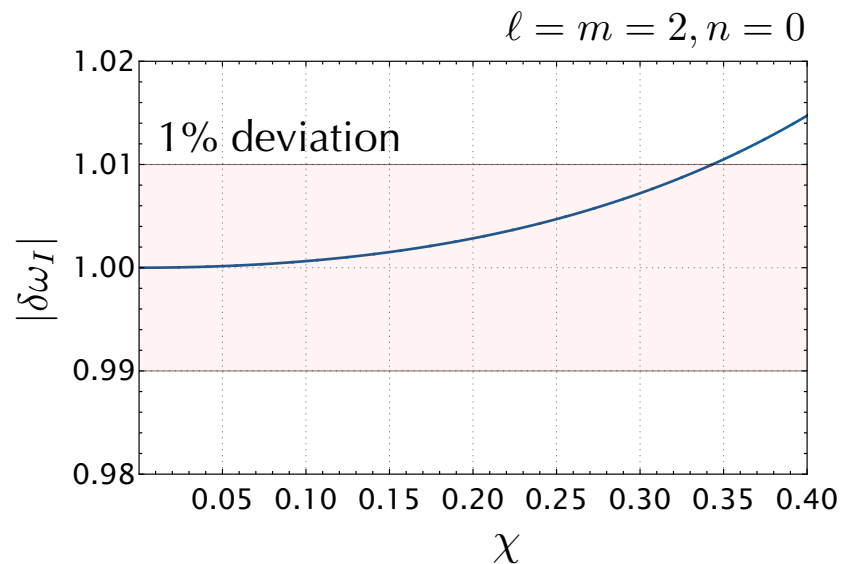
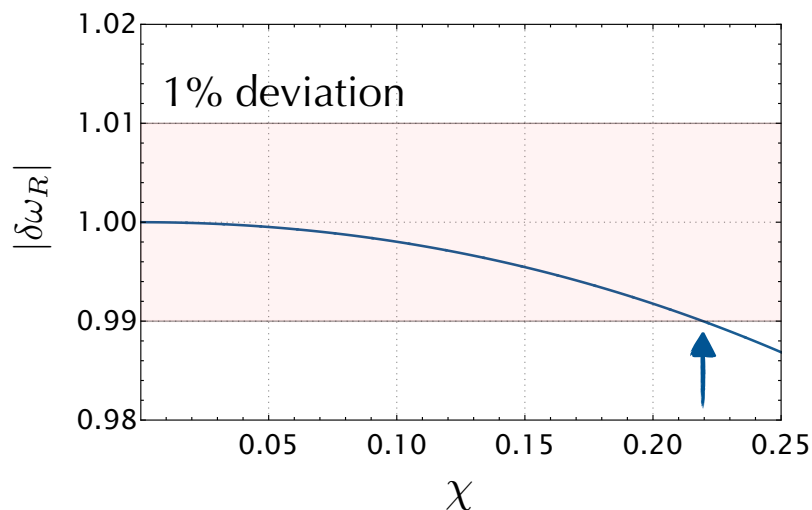
EM, Buoninfante, Mazumdar, Pani, PRD **102**, 064053 (2020)

Spinning horizonless compact objects

We generalize the membrane paradigm to spinning horizonless compact objects described by the **Kerr metric to linear order in spin**.

M.V.S. Saketh, EM, PRD **110**, 084038 (2024)

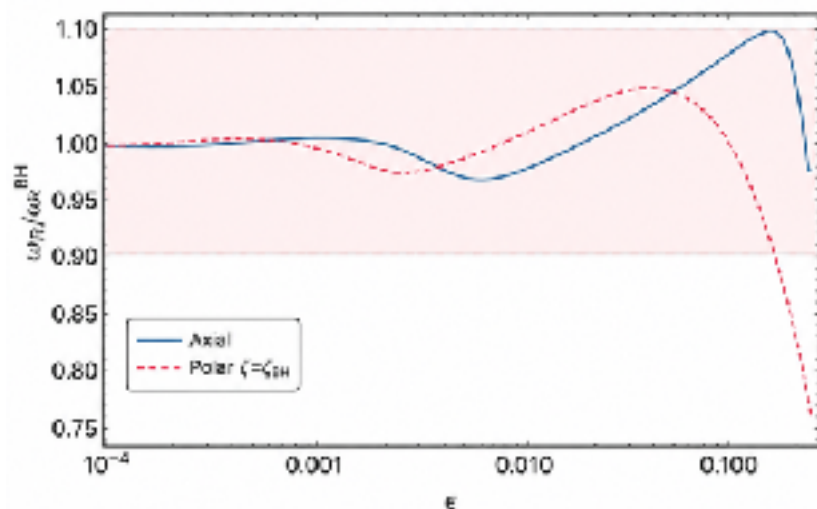
From a comparison with the black hole case, this description is valid until the maximum value of the spin $\chi = 0.2$.



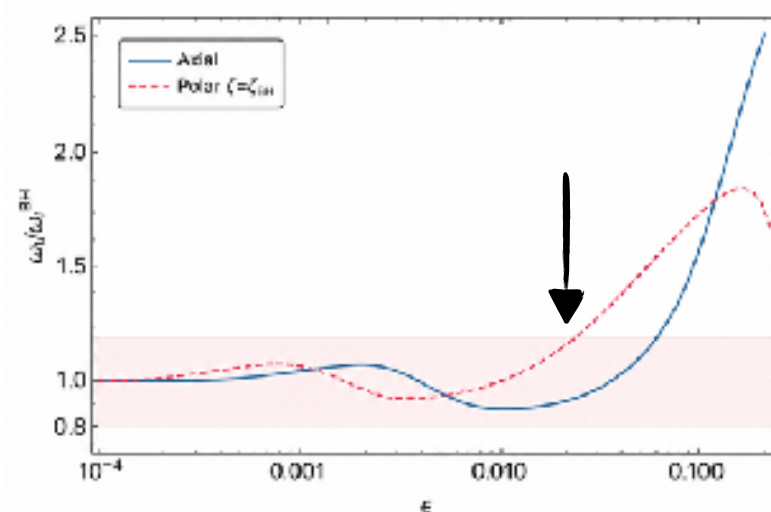
Quasi-normal mode spectrum

The spin makes the constraints on the compactness of objects more stringent.

$$\chi = 0.2$$



$$\ell = m = 2, n = 0$$



M.V.S. Saketh, EM, PRD **110**, 084038 (2024)

Horizonless compact objects with $\epsilon \lesssim 0.04$ are compatible with the measurement accuracy of the fundamental quasinormal mode of GW150914.

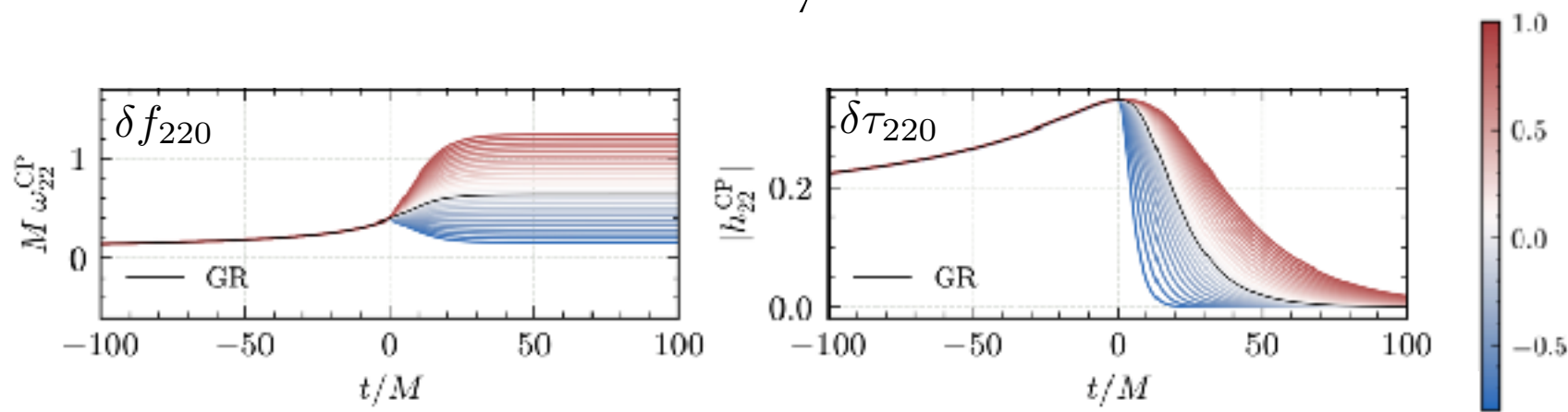
Parametrized ringdown test

We introduce fractional deviations to the frequency and the decay time of the fundamental quasinormal modes:

$$\omega_{\ell m 0} = \omega_{\ell m 0}^{\text{GR}}(1 + \delta\omega_{\ell m 0})$$

$$\tau_{\ell m 0} = \tau_{\ell m 0}^{\text{GR}}(1 + \delta\tau_{\ell m 0})$$

where the GR values are predicted from estimates of the mass and spin of the remnant and fits with numerical relativity.

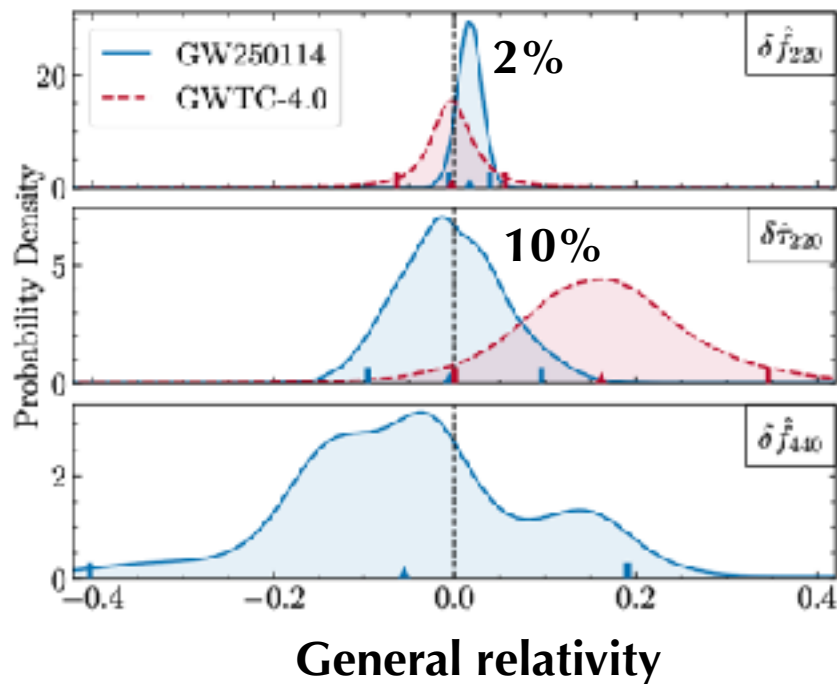


Ghosh+, PRD **103**, 124041 (2021); Brito+, PRD **98**, 084038 (2018); EM+, PRD **108**, 024043 (2023); Pompili+, PRD **111**, 124040 (2025)

GW250114

The clearest view of merging black holes (signal-to-noise ratio = 80)

Abac et al., PRL 135, 111403 (2025)

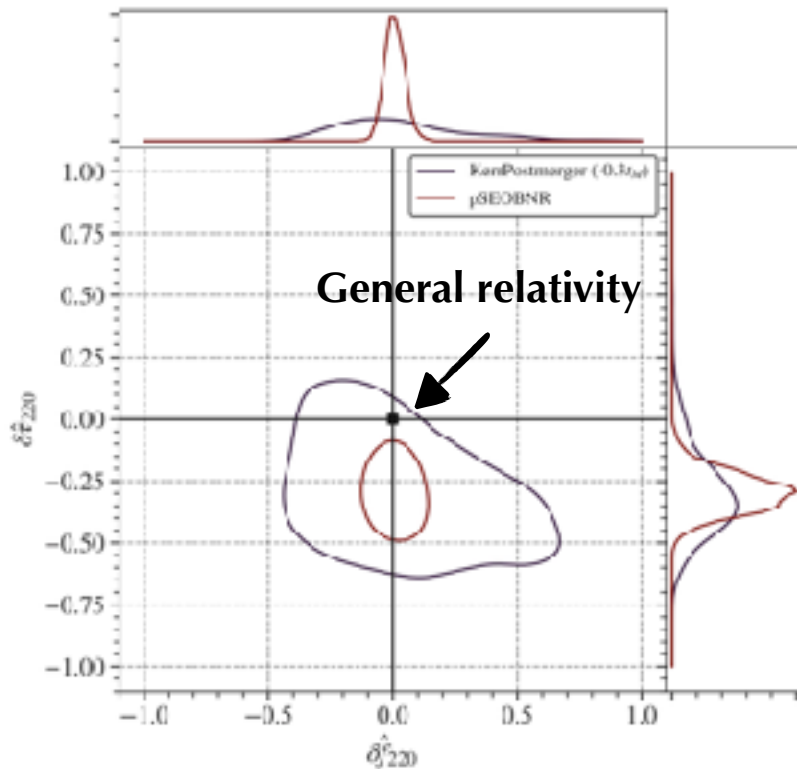


The fundamental mode is constrained twice as strictly as combining events from the fourth catalog.

Abac et al., arXiv:2509.08099; Grimaldi+, PRD **113**, L061506 (2026)

GW230814

The loudest GW in the fourth catalog detected by LIGO Livingston, with signal-to-noise ratio of 42.4.



The data supports a damping time smaller than expected from general relativity:

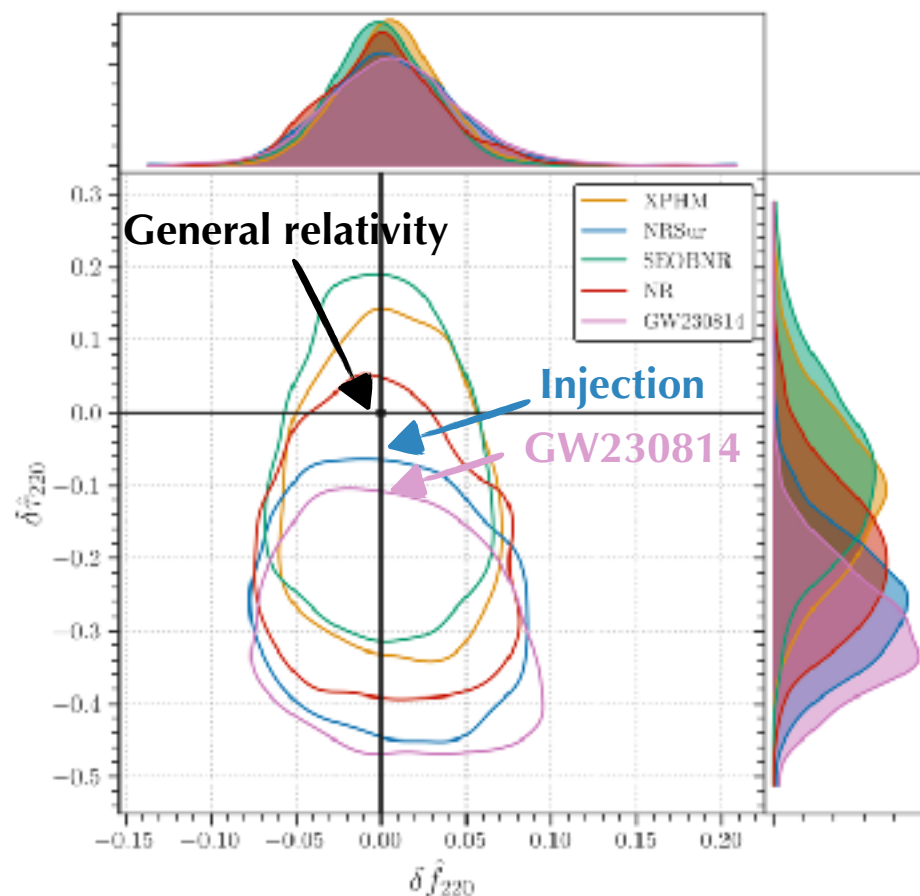
$$\delta\tau_{220} = -0.3^{+0.15}_{-0.12}$$

What causes this deviation?

Abac+, arXiv:2509.07348

Assessing waveform systematics

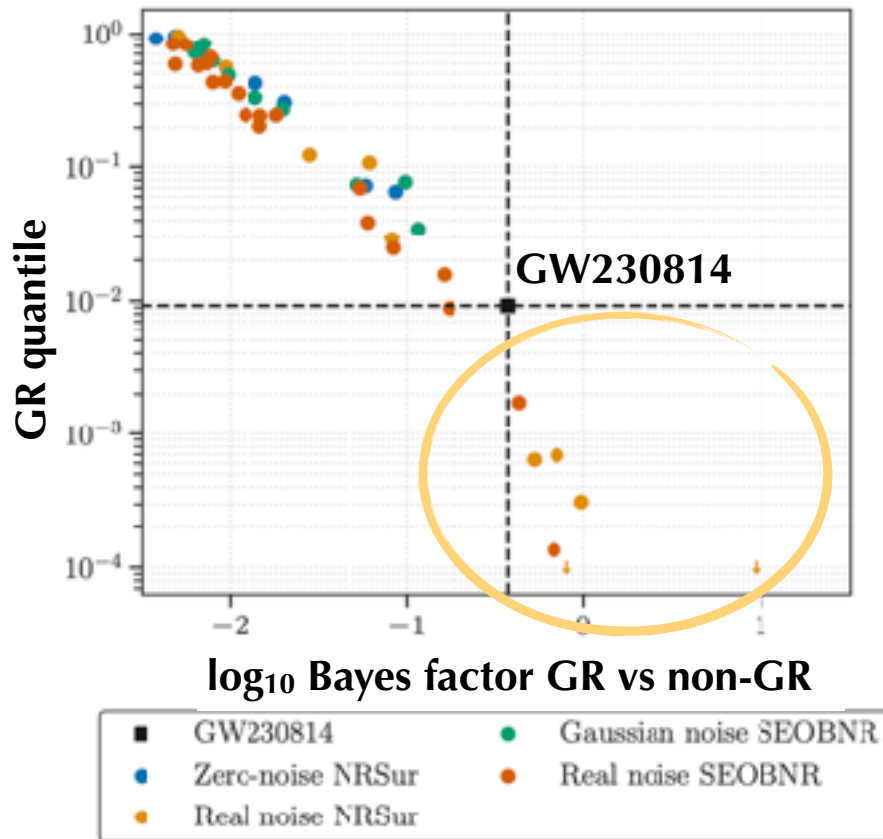
We perform injections with different waveform models in zero noise.



Waveform systematics could be responsible of the observed bias in the ringdown damping time.

Assessing detector effects

We perform injections in the real noise of the detector at various times around the event.



Statistical fluctuations in real detector noise can be responsible for apparent deviations from GR in the ringdown.

Detectability of echoes

- A tentative evidence for echoes in LIGO/Virgo data has been reported

Abedi+, PRD **96**, 082004 (2017); Conklin+, PRD **98**, 044021 (2018)

- Independent searches argued that the statistical significance of echoes is consistent with noise

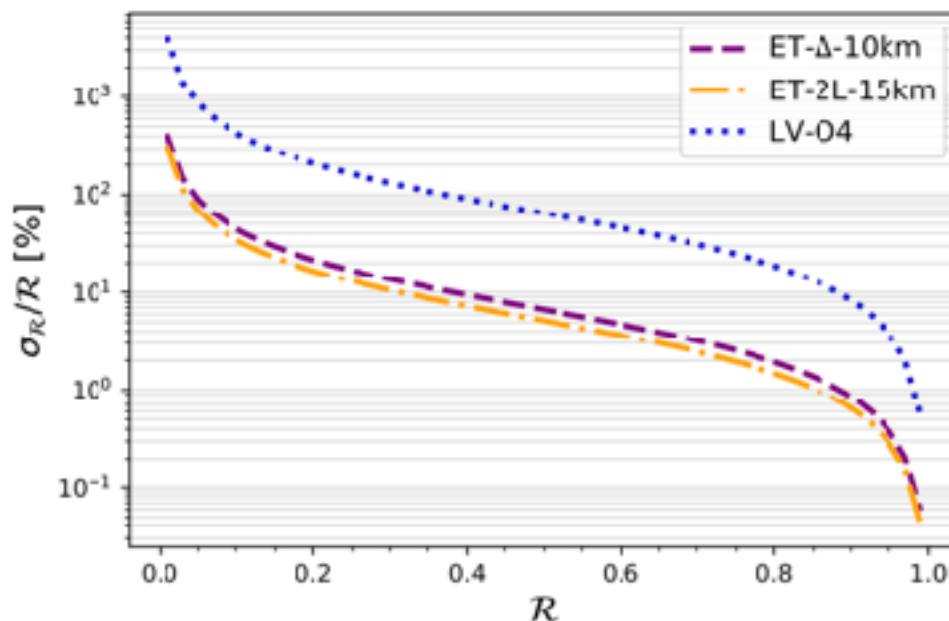
Westerweck+, PRD **97**, 124037 (2018); Nielsen+, PRD **99**, 104012 (2019); Uchikata+, PRD **100**, 062006 (2019); Lo+, PRD **99**, 084052 (2019); Tsang+, PRD **101**, 064012 (2020); Miani+, PRD **108**, 064018 (2023)

- No evidence for echoes by the LIGO/Virgo/KAGRA collaboration

Abbott+, PRD **103**, 122002 (2021); Abbott+, PRD **112**, 084080 (2021); Abac+, arXiv:2603.19019

Prospects with next-generation detectors

We can assess the **bounds on the reflectivity** of compact objects for a GW150914-like event. Branchesi+, JCAP **07** (2023) 068; Abac+, JCAP **03** (2026) 081



The accuracy on the reflectivity is affected by an order of magnitude between the O4 LIGO-Virgo and the **Einstein Telescope** configurations.

Conclusions and future prospects

- We can test the physics at the horizon scale with gravitational waves.
- Horizonless compact objects are not excluded by current observations.
- Next-generation detectors will allow us to perform unprecedented tests of the black hole paradigm.
- Improve the accuracy in the modeling of horizonless compact objects.
- Develop data analysis strategies to constrain the nature of the remnants.